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Medical SAM-Clip Grafting for brain tumor segmentation ☆

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ABSTRACT

Brain Tumor Segmentation (BTS) is crucial for accurate diagnosis and treatment planning, but existing CNN and Transformer-based methods often struggle with feature fusion and limited training data. While recent large-scale vision models like Segment Anything Model (SAM) and CLIP offer potential, SAM is trained on natural images, lacking medical domain knowledge, and its decoder struggles with accurate tumor segmentation. To address these challenges, we propose the **Medical SAM-Clip Grafting Network (MSCG)**, which introduces a novel SC-grafting module. This module grafts the rich semantic information from CLIP's feature space into the SAM model, enhancing its ability to capture both pixel-level details and high-level semantic context. By effectively combining SAM's visual precision with CLIP's semantic richness, we introduce SC-grafting module to improve the model's robustness and accuracy in brain tumor segmentation, especially in cases with irregular shapes, low contrast, or complex anatomical structures. Extensive experiments on the BraTS dataset demonstrate that MSCG significantly outperforms existing methods, showcasing its effectiveness in delivering more precise tumor segmentation and its potential in advancing medical image analysis.

1. Introduction

In the field of neurology, brain tumors like Glioblastoma can be lethal to the nervous system. These tumors exert pressure on the surrounding brain tissue, causing debilitating symptoms such as headaches, cognitive impairment, and poor concentration, ultimately impairing the patient's neurological function. The best treatment for brain tumors is surgical resection. However, due to the unresectable nature of normal brain tissues and the extensive infiltrative growth of malignant tumors, it is extremely difficult to achieve complete resection in surgical areas. With advancements in medical science, Magnetic Resonance Imaging (MRI) has become an invaluable tool for brain tumor analysis and diagnosis. As shown in Fig. 1, MRI utilizes multiple modalities to distinguish different tissues and areas, enabling the localization of the lesion and providing detailed information about the tumor to guide further medical treatment [1].

The treatment of brain tumors, especially Gliomas, requires fast and early diagnosis. Although human experts can provide accurate diagnoses, the conventional treatment process is often laborious and time-consuming. With the rapid growth of AI techniques in the medical field, Brain Tumor Segmentation (BTS) has emerged as a preliminary step in the diagnostic process. BTS can automatically locate and segment the brain tumor area in MRI images, which accelerates the treatment process and increases efficiency.

Techniques and methods for BTS are primarily linked to the field of computer vision, with most approaches capitalizing on Convolutional Neural Networks (CNNs) or Transformer models. Seminal CNN models like UNet [2] and UNet++ [3] have demonstrated their advantages in BTS tasks. These models excel at extracting and segmenting local details in the tumor area. However, due to their limited receptive fields, CNN-based methods struggle to gain a complete understanding of the entire MRI image, as human experts would, which is crucial for brain tumor

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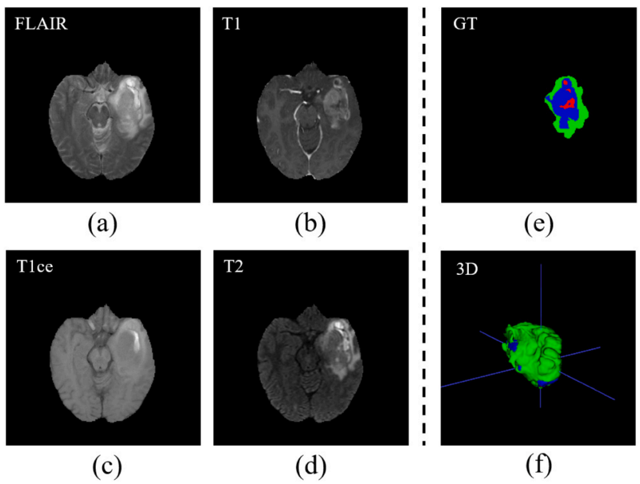

Fig. 1. An example of multimodal MRI for brain tumor segmentation is presented as follows: (a) The FLAIR modality; (b) The T1 modality; (c) The T1ce modality; (d) The T2 modality; (e) The ground truth (GT) segmentation label provided by domain experts; (f) 3D Volume.

diagnosis. In contrast, Transformer-based models like TransBTSV2 [4] and SwinBTS [5] utilize attention mechanisms, enabling the models to acquire a global receptive field. By considering the information from the entire MRI image, these Transformer-based methods can achieve better performance. Yet, Transformers divide images into patches for global information modeling, which can result in the loss of local detail and lead to blurred edge segmentation. Furthermore, the mainstream BTS datasets, such as BraTS [6], may be inadequate for effectively training Transformer-based models.

The Segment Anything Model (SAM) [7] has introduced new opportunities for the segmentation task. SAM has shown strong generalization abilities, which it achieved by utilizing one billion images for training. However, the images used for SAM's pre-training were primarily captured under natural conditions, which cannot provide SAM with medical-specific knowledge. As a result, further fine-tuning is required for SAM to perform well on the Brain Tumor Segmentation (BTS) task. Besides, the training of SAM focuses on the object's shapes and does not include any semantic prior knowledge. Thus, it focuses on the segmentation details and is hard to fine-tune on specific semantics. SAM also requires manually offered prompts like bounding boxes to generate desired segmentation results, which is counter to the goal of automation in the BraTS task.

Recently, the introduction of the CLIP (Contrastive Language-Image Pre-Training) model [8] has drawn researchers' attention. CLIP was trained on numerous image-text pairs for classification and has shown superior ability in extracting semantic information. To date, several groups have endeavored to adapt both CLIP and SAM for radiological tasks based on natural images, notably through the development of BiomedCLIP [9] and MedSAM [10], which were pre-trained on millions of biomedical datasets. However, these methods disregard the interaction between CLIP and SAM features.

In this paper, we propose the Medical SAM-Clip Grafting network (MSCG) for better brain tumor segmentation. Our method includes a Dual Encoder, a SC-grafting module, and a Fusion Decoder. The Dual Encoder uses SAM's encoder and CLIP's image encoder to parallelly extract geometric and semantic information. The SC-grafting module is used for feature alignment between the two branches, which enhances both features. The SC-grafting module aligns structural and semantic information between the two branches, enabling more complementary feature interaction and improved representation quality. Finally, a Fusion Decoder is employed for fusing the features from the two branches, using a Mamba layer for long-range information judgment. Our method is conducted on the BraTS2018 dataset, and the main contributions are as follows:

- We proposes the MSCG, which combines the advantages of both SAM and CLIP for better BraTS.
- We introduce a SC-grafting module, which grafts the rich semantic information from Clip's feature to SAM model, to help robust brain tumor modeling.
- We perform extensive experiments to verify the effectiveness of our proposed modules. Our approach achieves a new state-of-the-art performance on two BraTS datasets.

2. Related work

2.1. UNet-based methods

In medical segmentation, UNet [2] has achieved significant success in most challenges. Based on this fact, researchers have begun to design modules for improved feature extraction and fusion. Luu et al. [11] proposed using group normalization instead of batch normalization. They employed an asymmetrically large encoder and axial attention in the decoder to enhance feature extraction. Rehman et al. [12] utilized Residual Extended Skip (RES) connections to convert low-level features into middle-level features. They also designed a Wide Context (WC) block to increase the valid receptive field. Yazci et al. [13] proposed using the DMSF module and DMSU module to enhance segmentation capabilities. Features from the encoder branch are fused into the decoder branch using skip connections. Furthermore, feature fusion in CNNs, particularly when combining low-level details and high-level contextual information, remains challenging. Despite efforts like MFIB/UMB modules to improve this fusion, CNN-based methods still fall short compared to newer architectures like transformers, which are better suited for learning long-range dependencies and integrating diverse feature representations.

2.2. Multi-modal methods

Considering the multiple modalities (FLAIR, T1, T2) in the BRATS data, some researchers are attempting to aggregate multiple models to improve segmentation performance. Zhang et al. [14] proposed a Cross-Modality Feature Transition (CMFT) branch and a Cross-Modality Feature Fusion (CMFF) branch, which aim to learn rich feature representations by transferring knowledge across different modalities and fusing information from various modalities. Zhu et al. [15] sought to utilize edge information in multimodal MRI. They introduced an edge detection module and a feature fusion module, which includes a Multi-Feature Inference Block (MFIB) based on graph convolution for effective feature reasoning and information dissemination. Zhang et al. [16] proposed TMFormer, which employs a Uni-Modal Token Merging Block (UMB) focused on information extraction from individual modalities and a Multi-Modal Token Merging Block (MMB) responsible for fusing multiple modalities.

2.3. SAM-based methods

UNet-base and multi-modal methods have demonstrated their practicality on the BraTS task. However, they typically train on specific MRI datasets, which are often insufficient in quantity. The Segment Anything Model (SAM), trained on over one billion images, has shown great potential in multiple medical segmentation tasks.

One common way to adapt SAM for the medical domain is by adding adapters. Chen et al. [17] injected a series of 3D adapters into the transformer blocks of the image encoder, enabling the pre-trained 2D backbone to extract three-dimensional information from the input data. Wu et al. [18] introduced the Medical SAM Adapter (Med-SA), which incorporates domain-specific medical knowledge into

SAM using a lightweight yet effective adaptation technique. They also proposed Space-Depth Transpose (SD-Trans) to adapt 2D SAM to 3D medical images, as well as the Hyper-Prompting Adapter (HyP-Adpt) for prompt-conditioned adaptation. Another adaptation approach combines SAM with other fundamental models. Pandey et al. [19] proposed harnessing the capabilities of the YOLOv8 model for approximate bounding box detection across modalities, alongside SAM and High-Quality SAM (HQ-SAM) for fully automatic and precise segmentation. Peivandi et al. [20] enhanced SAM's mask decoder using transfer learning with the Decathlon brain tumor dataset, developing three methods to encapsulate four-dimensional data into three dimensions for SAM. Another line of research seeks to complement SAM with semantic information extracted from CLIP. Koleilat et al. [21] introduced the MedCLIP-SAM framework, which combines CLIP and SAM models to generate segmentation of clinical scans using text prompts in both zero-shot and weakly supervised settings. However, these methods often rely on *shallow or single-stage fusion*, and do not perform deep or scale-aware integration. In particular, they fail to model layer-wise feature interactions and lack mechanisms to align geometric and semantic spaces across multiple abstraction levels.

3. Method

Our MSCG is primarily composed of three parts: (1) a Dual Encoder, which consists of a SAM encoder and a CLIP image encoder to model brain tumor features from their pretraining weights; (2) a SC-adapting Module that grafts the encoding features from CLIP to SAM, which helps to obtain robust features; and (3) a Fusion Decoder that uses 3D convolution and a 3D Mamba layer to decode the encoder's features and generate accurate brain tumor segmentation results. In this section, we will further detail these components.

3.1. Preliminaries

To integrate Mamba into medical segmentation tasks, we first introduce the state-space modeling foundations that underlie the Mamba block. Mamba is based on the Structured State Space Sequence (S4) model, which originates from continuous-time linear time-invariant (LTI) systems. Such a system governs a one-dimensional input sequence $x(t) \in \mathbb{R}$ via a latent state $h(t) \in \mathbb{R}^N$, described by the following ordinary differential equation (ODE):

$$\frac{dh(t)}{dt} = Ah(t) + Bx(t), \quad y(t) = Ch(t) + Dx(t), \tag{1}$$

where $A \in \mathbb{R}^{N\times N}$, $B \in \mathbb{R}^{N\times 1}$, $C \in \mathbb{R}^{1\times N}$, and $D \in \mathbb{R}$. To apply this system to discrete sequences in deep learning, we employ the zero-order hold (ZOH) discretization scheme:

$$A_d = \exp(\Delta A), \quad B_d = A^{-1}(\exp(A) - I)\Delta B, \tag{2}$$

which leads to the discrete-time recurrence equations:

$$h_k = A_d h_{k-1} + B_d x_k, \quad y_k = Ch_k + Dx_k. \tag{3}$$

While this discretized form models long-range dependencies linearly, its dynamics are static across inputs. To overcome this limitation, Mamba introduces a *selective scan* mechanism, in which the matrices B, C, and Δ are dynamically predicted from the input data, making the system input-dependent and highly adaptive.

In our segmentation framework, we incorporate Mamba as 3D Mamba blocks within the decoder to model long-range interactions over volumetric features. Compared to transformer-based blocks, Mamba maintains linear computational complexity with respect to sequence length, making it more efficient for high-resolution 3D medical image segmentation.

3.2. Dual encoder

The BraTS data contains four MRI modalities of the brain: T1, T1-Gd, T2, and FLAIR. All four MRI modalities are essential for accurately segmenting different tumor regions, as they depend on the presence of vascularity and fluids within the brain. Thus, our Dual Encoder takes all four modalities as inputs. The original SAM and CLIP models are pretrained on natural images, which are not suitable for MRI images. We use the SAM-Med3D [22] image encoder and the CT-CLIP [23] image encoder to compose our Dual Encoder. However, both of these models are pretrained on single-modality inputs. To adapt multiple modalities to the pretrained model, we made minor modifications to the input images' embeddings.

As shown in the upper part of Fig. 2. Given an 3D MRI image $\mathbf{I}_{mri} \in \mathbb{R}^{4\times D\times H\times W}$, which contains $\mathbf{I}_{t1}$, $\mathbf{I}_{t1Gd}$, $\mathbf{I}_{t2}$, $\mathbf{I}_{flair}$. For SAM encoder, each modal image $\mathbf{I}_{mod} \in \mathbb{R}^{1\times D\times H\times W}$ will be encoded using the same Patch Embedding layer. Then, the 4 embedded feature will be merged by using a linear layer to the multi-modality feature $\mathbf{E}_{SAM} \in \mathbb{R}^{4\times \frac{D}{16}\times\frac{H}{16}\times\frac{W}{16}}$,

$$\mathbf{E}_{mod} = \psi(\mathbf{I}_{mod}), \quad mod = t1, t1gd, t2, flair, \tag{4}$$

$$\mathbf{E}_{SAM} = \Gamma([\mathbf{I}_{t1}, \mathbf{I}_{t1Gd}, \mathbf{I}_{t2}, \mathbf{I}_{flair}]), \tag{5}$$

where ϕ donates the Patch Embedding layer, Γ represents the Linear layer and $[\cdot]$ donates the concatenation operation. Eqs. (4) and (5) define the two branches used for feature extraction in our framework. In Eq. (4), each individual MRI modality (i.e., T1, T1-gd, T2, and FLAIR) is independently encoded through a shared feature extractor $\psi(\cdot)$ to obtain modality-specific features $\mathbf{E}_{mod}$. This allows the model to preserve fine-grained structural details within each modality. In contrast, Eq. (5) describes the second branch, where all four modalities are concatenated along the channel dimension and jointly processed by the pre-trained SAM encoder $\Gamma(\cdot)$ to produce a fused semantic feature $\mathbf{E}_{SAM}$. This branch focuses on learning global contextual representations that integrate multi-modality cues.

Unlike the SAM part, CLIP encoder will embed the 4 model images using 4 separately randomly initialized Patch Embedding layers. These embedded features will be merged by linear layers to the multi-modality feature $\mathbf{E}_{clip} \in \mathbb{R}^{4\times \frac{D}{16}\times\frac{H}{16}\times\frac{W}{16}}$. CLIP is structure-oriented, extracting fine-grained features from each modality independently, whereas SAM is semantics-oriented, encoding all modalities together to learn holistic contextual information. The two different embedding method made the SAM model focus on the tumor's edges and shape while CLIP model focus on the tumor's semantic across the different MRI modalities.

After that, the embedding feature $\mathbf{E}_{clip}$ will be sent into the pre-trained CLIP model to gain the encoded CLIP feature $\mathbf{B}_{clip}$. And $\mathbf{E}_{SAM}$ will goes into the Transformer with loaded pre-trained weight to generate the encoded features,

$$\mathbf{A}_i = STrans_i(\mathbf{A}_{i-1}), \quad i = 1, \ldots, 12, \tag{6}$$

where $STrans_i$ donates the ith Transformer blocks in the SAM model, A_i donates the corresponding layers features.

To overcome the lack of brain tumor semantic information in SAM's features, Dual Encoder uses SC-grafting module to guided the SAM feature by clip feature $\mathbf{B}_{clip}$,

$$\mathbf{F}_t = Graft_t(\mathbf{A}_i, \mathbf{B}_{clip}), \quad i = 3, 6, 9, 12 \tag{7}$$

where $Graft_i$ represents the SC-grafting module, $\mathbf{F}_t$ donates the output features. The detailed operations of the grafting process are illustrated in Fig. 3. A step-by-step explanation will be provided in Section 3.4. The output features will later be used in the Fusion Decoder. The output features will later be used in the Fusion Decoder.

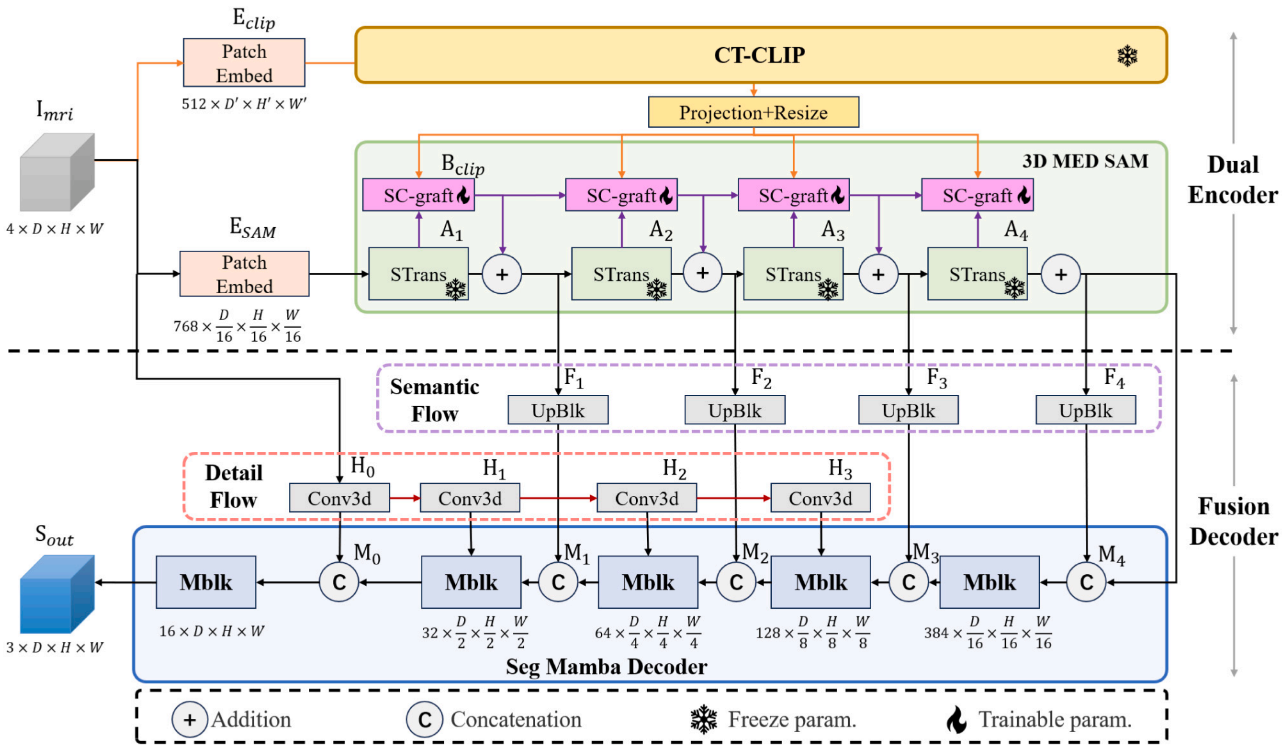

Fig. 2. Overview of the MSCG architecture. The Dual Encoder combines SAM and CLIP encoders to extract structural and semantic features. SC-grafting modules align these features at multiple transformer layers. The Fusion Decoder integrates Semantic Flow, Detail Flow, and 3D Mamba Blocks to generate high-resolution segmentation outputs.

3.3. Fusion decoder

While the transformer architectures in the Dual Encoder can effectively extract and model brain tumor information, they are often difficult to implement in the decoder due to their significant computational burden when handling excessively long feature sequences. Most brain tumor segmentation methods, like UNetR and UNet++, use 3D convolutional layers as decoders because of their computational efficiency. However, their limited receptive field restricts the decoding of multi-scale features, which are vital for accurate segmentation results. To address this issue, we introduce the Fusion Decoder to effectively engage the features from the Dual Encoder. As shown in the lower part of Fig. 2, our Fusion Decoder is composed of Semantic Flow, Detail Flow, and Seg Mamba Decoder.

Semantic Flow. As the purple dot box shown in Fig. 2, for the Dual Encoder output features $\mathbf{F}_t$ where $t \in [1, 2, 3, 4]$, the Semantic Flow will transform and reshape these patch-wise feature into 3D feature shape with original dimension,

$$\widetilde{\mathbf{F}}_t = \Upsilon(\mathbf{F}_t) \tag{8}$$

where $\widetilde{\mathbf{F}}_t$ is the 3D feature with dimension $\mathbb{R}^{2^{5+i} \times \frac{D}{2^i} \times \frac{H}{2^i} \times \frac{W}{2^i}}$. Υ donates the Upblk, which repeatably using upsample and 3d convolutional layers to transform feature to the desired dimension.

Detail Flow. The patch-wise encoder feature $\mathbf{F}_t$ lacks of the high-resolution detail, such as the tumor's accurate edge. Thus, we proposed a Detail Flow in the Fusion Decoder. As shown by the red dotted box in Fig. 2, for input image $\mathbf{I}_{mri}$, the Detail Flow will use 3d convolutional layers and downsampling to generate a series of high-resolution features $\mathbf{H}_t$,

$$\mathbf{H}_0 = f_{3d}(\mathbf{I}_{mri}), \tag{9}$$

$$\mathbf{H}_t = f_{3d}(\daleth(\mathbf{H}_{t-1})), \quad i = 1, 2, 3 \tag{10}$$

where f_{3d} is the 3d convolutional layer with InstanceNorm and GELU activation, $\daleth$ donates the 2x down sampling layer.

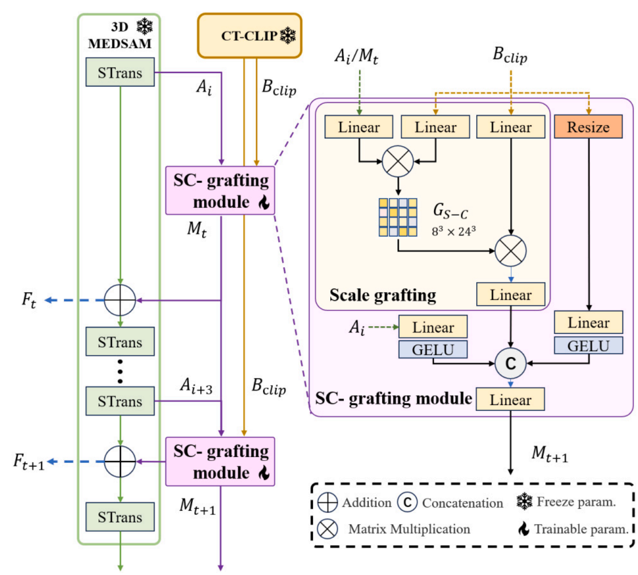

Fig. 3. The detailed structure of SC-grafting module.

Seg Mamba Decoder. The projected semantic features $\widetilde{\mathbf{F}}_t$ and detail features $\mathbf{H}_t$ need to be adequately fused. While the convolutional layers are limited by their receptive field, transformer layers require high computational resources. Deriving benefit from the selective scan operation, the Mamba block can achieve a global receptive field with a linear computational resource cost. Hence, we adopt 3D Mamba [24] to fuse the semantic and detail features.

As shown by the blue box in Fig. 2, the Seg Mamba Decoder is composed of several Mamba Blocks, denoted as Mblk. Each Mblk takes the previous Mblk's features $\mathbf{M}_{t+1}$, and concatenate with the corresponding semantic features $\mathbf{F}_{t+1}$ as the basic feature. As indicated

Table 1
Quantitative comparison of different brain tumor segmentation methods on the BraTS2023 benchmark.

Method	WT		TC		ET		Average	
	Dice	HD95	Dice	HD95	Dice	HD95	Dice	HD95
U-Net++	87.70	7.34	85.10	6.52	78.25	4.31	83.68	6.06
MedNeXt	90.02	5.07	87.10	5.08	81.66	4.88	87.26	5.01
SegresNet	90.41	5.98	85.75	5.67	81.96	5.51	86.04	5.72
TransUnet	89.25	5.59	86.19	5.47	80.41	4.37	85.28	5.14
TransBTS	88.06	6.56	79.30	6.78	76.36	4.84	81.24	6.06
SwinBTS	89.00	6.64	80.94	7.04	77.93	4.73	82.62	6.14
UNETR	89.71	7.22	87.79	4.42	83.48	5.52	86.99	5.72
SwinUNETR	90.19	7.17	86.39	5.29	81.49	5.03	86.02	5.83
SegMamba	90.52	7.52	89.57	4.86	81.55	4.97	87.21	5.78
SAM	87.66	13.98	84.88	9.67	76.47	8.40	83.01	10.68
MedSAM	89.46	6.11	86.75	5.79	81.22	5.74	85.81	5.88
SAM2	88.95	7.45	86.13	6.30	80.31	5.72	85.13	6.49
MSCG	90.66 ± 0.31	7.73 ± 0.07	89.03 ± 0.27	4.51 ± 0.04	82.47 ± 0.22	4.80 ± 0.04	87.39 ± 0.24	5.68 ± 0.05

by the red arrow in Fig. 2, each Mblk also takes the detail features $\mathbf{H}_t$ as complementary high-resolution detail information. Each Mblk will generate segmentation results $\mathbf{S}_t$ for deep supervision. Thus, the operation at the ith Mblk can be formulated as follows:

$$\mathbf{M}_t = Mblk([\mathbf{M}_{t+1}, \mathbf{F}_{t+1}] + \mathbf{H}_t), \tag{11}$$

$$\mathbf{S}_t = \delta(\Phi(\mathbf{M}_t)), \tag{12}$$

where δ refers to the Sigmoid activation function, Φ is the 1×1 3d convolutional layer. Under the Seg Mamba Decoder, the semantic and detail information can be efficiently merged to generate high-quality predictions.

3.4. SC-grafting module

The SAM model is pretrained to segment images without modeling semantic information, which lacks the capability to capture brain tumor-related features. While the CLIP model is pretrained to align with text prompts, the CLIP encoder feature $\mathbf{B}_{clip}$ is a better option for brain tumor semantic modeling. Directly concatenating the two modules leads to suboptimal results. Thus, we propose the SC-grafting module, which contains a scale grafting operation. It can graft the brain tumor semantic feature from the CLIP encoder feature $\mathbf{B}_{clip}$ to SAM encoder features $\mathbf{A}_i$, gaining advantages from both pretrained models.

Scale grafting. As shown in Fig. 3, the SC-grafting module takes the CLIP encoder feature $\mathbf{B}_{clip} \in \mathbb{R}^{512 \times L2}$ and the $\mathbf{A}_i$ or $\mathbf{M}_t \in \mathbb{R}^{768 \times L1}$ as inputs. Note that in our MSCG, $L1 = 8^3$ and $L2 = 12^3$. We unify their different resolutions by using Scale grafting. Specifically, the two features will be project to a smaller dimension, and then a scale grafting map $\mathbf{G}_{S-C}$ is generated by matrix multiplication, which can be represented as follows:

$$\hat{\mathbf{A}}_i = W_q(\mathbf{A}_i), \tag{13}$$

$$\hat{\mathbf{B}}_{clip} = W_k(\mathbf{B}_{clip}), \tag{14}$$

$$\mathbf{G}_{S-C} = \xi(\hat{\mathbf{A}}_i \times \hat{\mathbf{B}}_{clip}) \tag{15}$$

where W_q and W_k are linear layers, $\times$ donates matrix multiplication and ξ refers to Softmax function. Scale grafting map $\mathbf{G}_{S-C}$ can be used to project clip features to SAM's feature dimension. Formally,

$$\mathbf{M}_{temp} = \mathbf{G}_{S-C} \times W_v(\mathbf{B}_{clip}), \tag{16}$$

where W_v is a linear projection and $\mathbf{M}_{temp}$ is the projected clip features.

Scale grafting can graft the CLIP feature to the SAM from different resolutions. However, semantic information is impaired due to the dimension shrinkage. Thus, we directly resize its resolution without drastically altering its dimension. Both features are projected to a feature with $\mathbb{R}^{384 \times L}$ dimension by fully-connected layers, which can be expressed as:

$$\widetilde{\mathbf{A}}_i = \sigma(W_{fc1}(\mathbf{A}_i)), \tag{17}$$

$$\widetilde{\mathbf{B}}_{clip} = \sigma(W_{fc2}(\mathbf{B}_{clip})), \tag{18}$$

where W_{fc1} and W_{fc2} are fully-connected layers, σ donates the GELU activation function. Then, all features will be merged by another fully-connected layer to generate the grafted feature,

$$\mathbf{M}_t = \sigma(W_{fc3}([\widetilde{\mathbf{A}}_i, \widetilde{\mathbf{B}}_{clip}, \mathbf{M}_{temp}])), \tag{19}$$

$$\mathbf{F}_t = \mathbf{M}_t + \mathbf{A}_t \tag{20}$$

where W_{fc3} is a fully-connected layer and t refers to t-th SC grafting module. The Dual encoder features $\mathbf{F}_t$ will be send to Fusion Decoder for prediction. And grafted feature $\mathbf{M}_t$ will be used fro next SC-grafting module.

Considering computational efficiency, we applied the SC-grafting module to the $i \in [3, 6, 9, 12]$ -th layer to enhance the SAM encoder feature across different scales (see Fig. 4).

3.5. Loss function

Our proposed MSCG is trained using three loss functions, namely Dice loss [25], BCE loss [26], IoU loss [27], and Focal Tversky Loss [28]. We denote these four losses as $\mathcal{L}_{dice}$, $\mathcal{L}_{bce}$, $\mathcal{L}_{iou}$ and $\mathcal{L}_{focal}$.

The BCE loss and IoU loss are widely adopt by all kinds of segmentation tasks, which can be defined as:

$$\mathcal{L}_{bce} = -y log(\hat{y}) - (1 - y) log(1 - \hat{y}) \tag{21}$$

$$\mathcal{L}_{iou} = 1 - \frac{y * \hat{y}}{y + \hat{y}} \tag{22}$$

where y represents the segmentation results and $\hat{y}$ donates the ground-truth.

Dice Loss is a metric that measures the similarity between samples and is commonly used in semantic segmentation tasks, particularly when addressing the issue of foreground-background imbalance, which can be defined as follows:

$$\mathcal{L}_{dice} = 1 - \frac{\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i} \tag{23}$$

where y_i represents the ith pixel in the segmentation prediction.

By focusing more on the hard-to-classify examples, FocalTversky-Loss improves model performance on minority classes, making it a powerful choice for tasks where segmentation accuracy is critical, which can be defined as:

$$\mathcal{L}_{focal} = \frac{1 - T_\alpha}{(1 - T_\alpha)^\gamma} \tag{24}$$

$$T_\alpha = \frac{|y \cap \hat{y}|}{|y \cap \hat{y}| + \alpha |y - \hat{y}| + (1 - \alpha)|\hat{y} - y|} \tag{25}$$

where T_α is the TverskyLoss, α and γ are focal parameters.

Each Mamba block in the Seg Mamba Decoder will generate a brain tumor segmentation $\mathbf{S}_t$. All of predictions at different scales will be

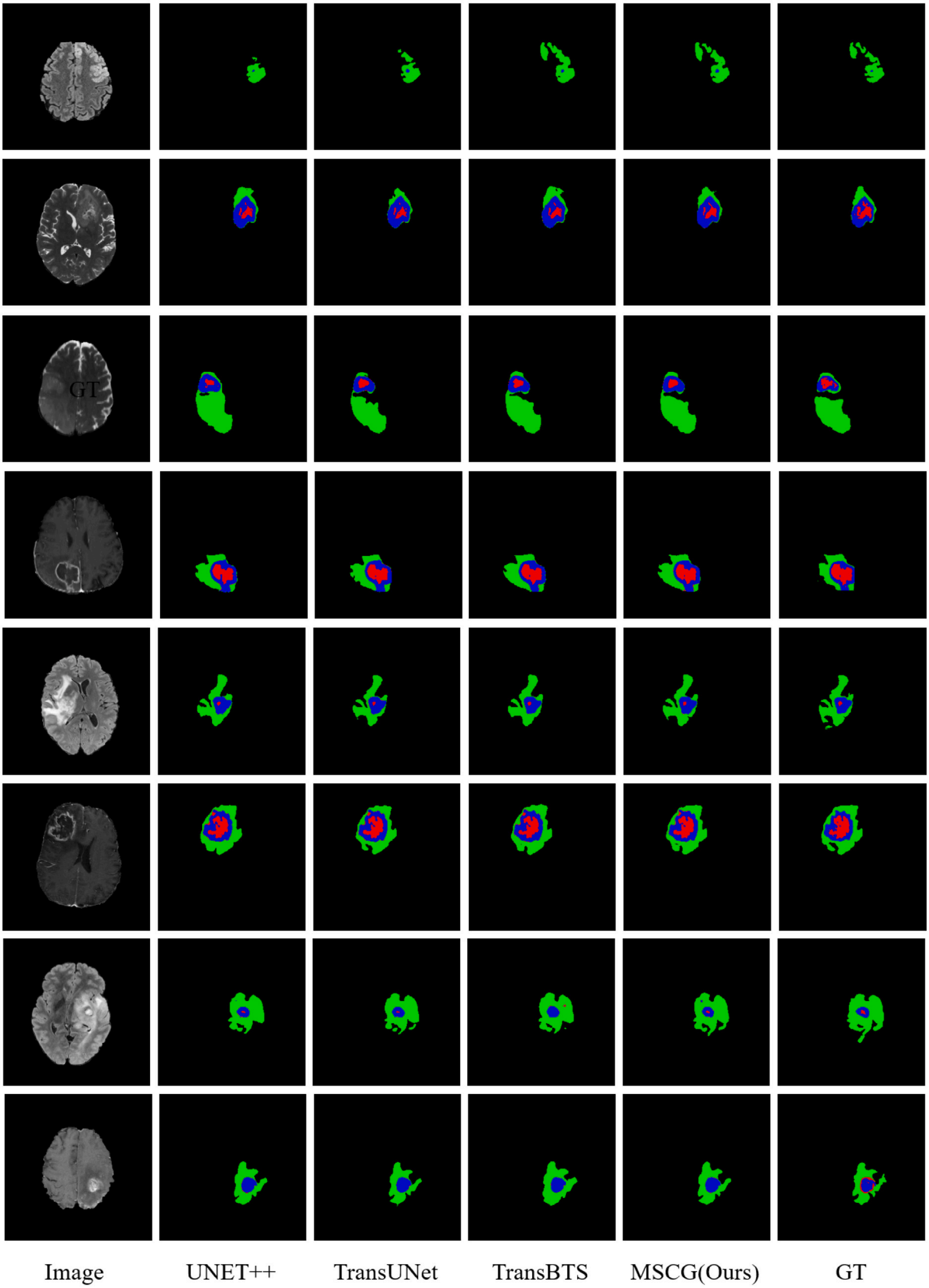

Fig. 4. Visual comparison of brain tumor segmentation results on BraTS2023 obtained by different methods. The green, blue and red indicate ED, ET and NCR/NET regions, respectively.

Table 2
Quantitative comparison of different brain tumor segmentation methods on the BraTS2018 benchmark.

Method	WT		TC		ET		Average	
	Dice	HD95	Dice	HD95	Dice	HD95	Dice	HD95
U-Net++	88.96	5.32	84.65	8.53	79.49	4.29	84.36	6.04
SwinBTS	90.02	5.07	87.10	5.08	81.66	4.88	86.26	5.01
TransUnet	90.25	4.39	87.19	5.54	80.41	3.73	85.95	4.55
UNETR	90.55	4.48	87.09	5.53	81.40	3.12	86.34	4.37
SegMamba	90.40	4.79	88.90	5.29	81.76	3.81	87.02	4.63
SAM	88.45	8.56	84.55	12.03	79.33	8.72	84.11	9.77
MedSAM	89.46	4.85	86.75	6.07	81.22	3.90	85.98	4.94
SAM2	89.03	4.89	86.28	6.20	81.04	4.06	85.45	5.05
MSCG	91.53 ± 0.28	4.68 ± 0.09	89.12 ± 0.16	5.87 ± 0.04	80.94 ± 0.24	3.97 ± 0.05	87.19 ± 0.21	4.84 ± 0.05

supervised with resized ground truth. The overall loss $\mathcal{L}_{all}$ for the MSCG can be represented as:

$$\mathcal{L}_{out} = \mathcal{L}_{dice} + \mathcal{L}_{bce} + \mathcal{L}_{iou} + \mathcal{L}_{focal}, \tag{26}$$

$$\mathcal{L}_{all} = \sum_{i=1} (\mathcal{L}_{out}(\mathbf{S}_t, \mathbf{S}_{gt})) \tag{27}$$

where $\mathbf{S}_{gt}$ represents the segmentation ground truth.

4. Experiment

4.1. Datasets

In the experiments, both the training and testing datasets are based on the BraTS2023 [6,29,30] benchmarks. As a vital public dataset for multimodal brain tumor segmentation, BraTS is utilized in the annual MICCAI brain tumor segmentation challenge and is widely adopted in related research. Each year's competition sees additions, deletions, or replacements in the dataset to enhance its scale. Specifically, BraTS2023 contains 1,251 annotated brain tumor samples for model training.

Each case includes MRI scans across four different modalities (FLAIR, T1, T1ce, and T2), all labeled by domain experts. The labels represent four classes: background, NCR/NET, ED, and ET. Evaluation is conducted based on three distinct brain tumor regions: Whole Tumor (WT = NCR/NET + ED + ET), Tumor Core (TC = NCR/NET + ET), and Enhancing Tumor (ET). In this paper, we employ two commonly used metrics in medical image segmentation for performance assessment: the Dice Score and the 95% Hausdorff Distance (HD).

4.2. Implementation detail

In the preprocessing stage, each scan measures $\mathbb{R}^{240\times240\times155}$. In this paper, we applied random flip and random corp to transform the scan volume into a uniform dimension $\mathbb{R}^{128\times128\times128}$. Both the SAM model and CLIP model take all four modalities as input. Additionally, popular Z-score normalization is applied to the raw data to address inconsistencies in image contrast across different modalities.

Our objective is to showcase the potential of SAM and CLIP in the brain tumor field. Thus, we freeze the parameters of CLIP's and SAM's image encoders, adapting SAM to the medical field. We train our method end-to-end using Stochastic Gradient Descent (SGD), with a learning rate of 0.01 for the other trainable parts. The maximum number of epochs is set to 200, and the batch size is set to 2. We also use cosine annealing as the learning rate decay scheduler.

The experimental results were conducted using an Intel Core i7 processor with an NVIDIA 4090D GPU, utilizing Pytorch [31]. The dataset was then split into training (70%) test (20%) and val(10%) sets for the validation of each case.

4.3. Comparison with methods

Quantitative Comparison As shown in Table 1 and Table 2, we compare our MSCG with other state-of-the-art brain tumor segmentation methods, which include three CNN-based approaches: U-Net++ [3], MedNeXt [32], and SegresNet [33], as well as a Mamba-based method, SegMamba [24]. The other methods are transformer-based: TransUnet [34], TransBTS [35], SwinBTS [5], UNETR [36], and SwinUNETR [37]. Additionally, we compare with SAM-based methods: SAM [7], MedSAM [38] and SAM2 [39]. For a fair comparison, we utilize public implementations and the given pretrained weights of these methods to infer under the same settings.

For quantitative evaluation on the BraTS2023 dataset, we adopt the Dice score (Dice) and 95% Hausdorff Distance (HD95). We use the official evaluation codes from performance computation. The segmentation results of brain tumors for the BraTS2023 dataset are presented in Table 2. Our MSCG achieves Dice scores of 90.66%, 89.09%, and 82.47%, along with HD95 values of 7.73%, 4.51%, and 4.80% for the WT, TC, and ET categories, respectively. These results indicate that MSCG exhibits superior segmentation robustness among the finetuning models.

Visual Comparison To compare the segmentation results of different methods more intuitively, we selected six comparative methods for visual analysis across the BraTS datasets. As shown in Fig. 4, our MSCG accurately detects the boundaries of each tumor region in the BraTS2023 dataset, demonstrating better consistency compared to other state-of-the-art approaches.

4.4. Ablation study

Effectiveness of key components Our MSCG contains four key components: Dual Encoder (DE), Detail Flow (DF), Semantic Flow (SF), and the SC-grafting module (SCG). In Table 3 and Fig. 6, we show the ablation results of using different key components. Results 1–2 in Table 3 suggest that combining the CLIP model's semantic information with the SAM features will produce better outcomes. Result 5 demonstrates better performance than Result 4, indicating that it is more effective to add high-resolution detail information. In Fig. 5, we show the segmentation results of using different components. One might observe that the CLIP model and Semantic Flow can help locate the accurate position of the tumor area, while Detail Flow can restore the precise segmentation boundary.

Effectiveness of using different loss function To demonstrate the necessity and effectiveness of our multi-task loss formulation, we conduct an ablation study by progressively introducing additional loss components on top of the base BCE loss, as summarized in Table 6. Starting from BCE alone, we successively add Dice, IoU, and Focal Tversky losses. The performance consistently improves across all metrics, indicating that the combined optimization of these losses does not introduce conflict but rather contributes complementary benefits.

Specifically, Dice loss emphasizes shape-level supervision, IoU introduces global overlap constraints, and Focal Tversky loss helps focus

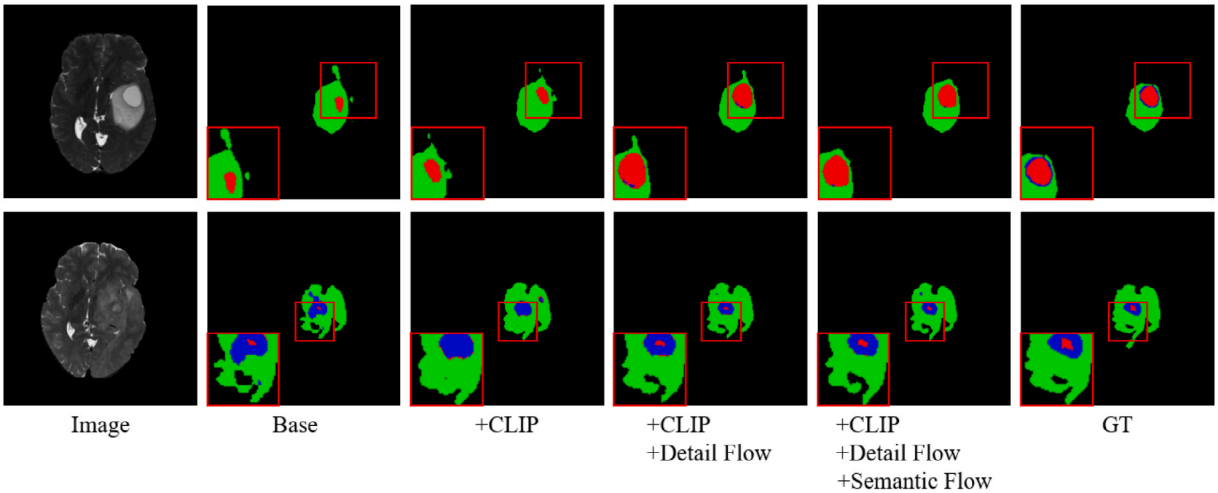

Fig. 5. Visual comparison of brain tumor segmentation results on BraTS2023 obtained by different key components in MSCG. The green, blue and red indicate ED, ET and NCR/NET regions, respectively.

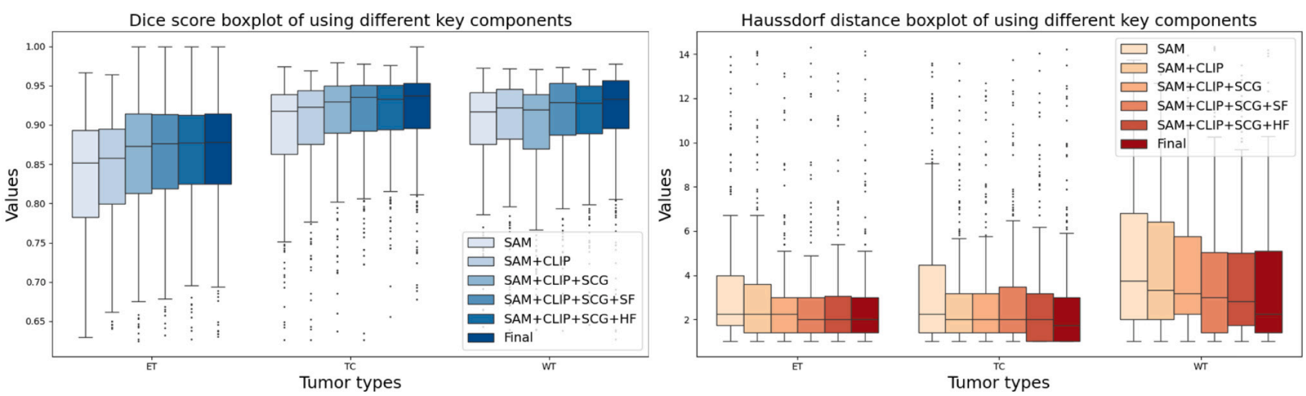

Fig. 6. Dice scores and Haussdorf distance boxplots of using different key components in MSCG.

Table 3
Quantitative comparison of key components in the proposed MSCG framework.

	Components					WT		TC		ET		Ave.	
	SAM	CLIP	SCG	DF	SF	Dice	HD95	Dice	HD95	Dice	HD95	Dice	HD95
1	✓	–	–	–	–	87.66	13.98	84.88	9.67	76.47	8.40	83.01	10.68
2	✓	✓	–	–	–	88.36	12.17	85.55	6.95	78.51	7.26	84.14	8.79
3	✓	✓	✓	–	–	87.87	9.29	87.48	5.83	81.09	6.06	85.46	7.06
4	✓	✓	✓	–	✓	89.63	9.49	87.92	5.04	81.74	5.37	86.43	6.63
5	✓	✓	✓	✓	–	89.98	8.11	88.50	4.80	81.73	4.53	86.73	5.81
6	✓	✓	✓	✓	✓	90.66	7.73	89.03	4.51	82.47	4.80	87.39	5.68

Table 4
Comparison of using different decoder architectures.

Architecture	WT		TC		ET		Ave.	
	Dice	HD95	Dice	HD95	Dice	HD95	Dice	HD95
Conv Blocks	89.89	6.62	85.91	4.96	76.90	6.86	84.23	6.14
Swin Blocks	91.37	8.04	87.61	4.74	80.77	5.88	86.58	6.22
Mamba Blocks	90.66	7.73	89.03	4.51	82.47	4.80	87.39	5.68

on hard-to-segment regions and boundary refinements. These results validate the necessity of our composite loss design and suggest that multi-task optimization enhances overall segmentation quality.

Effectiveness of using Mamba layer To better demonstrate the advantages of using the Mamba structure as a decoder component, we replace the positions of Mamba blocks in the Fusion Decoder with several major network structures and conduct experiments. The results are shown in Table 4 and Fig. 7. Among them, the window sizes of the Swin Block are set to 16, respectively. It can be observed that, while using a similar computational load, the Mamba block achieves better results. This indicates that Mamba can interact with distant locations, giving it a larger receptive field compared to convolutional neural networks. Additionally, for a given position, it is more sensitive to geometrically nearby regions, making it easier to model local structural information than Transformer networks. Therefore, using Mamba blocks is a more reasonable choice for predicting brain tumor areas with better detail.

Effectiveness of using different trainable parameter. Considering that the weights of the SAM and CLIP models are pretrained on

Table 5
Comparison of using different trainable parameter.

Training config	WT		TC		ET		Ave.	
	Dice	HD95	Dice	HD95	Dice	HD95	Dice	HD95
SAM	90.49	7.88	87.72	4.39	80.04	4.89	86.08	5.72
SAM+adapter	90.65	7.86	88.23	4.40	81.43	4.87	86.77	5.71
CLIP	89.22	8.72	84.44	7.40	74.78	8.73	82.81	8.28
CLIP+adapter	89.56	8.50	84.55	7.24	75.28	8.38	83.13	8.04
SC-grafting	90.66	7.73	89.03	4.51	82.47	4.80	87.39	5.68
SAM+CLIP+SC-grafting	90.42	7.94	88.39	4.86	81.82	4.56	86.87	5.78

Table 6
Comparison of using different loss function.

Loss function	WT		TC		ET		Ave.	
	Dice	HD95	Dice	HD95	Dice	HD95	Dice	HD95
BCE	89.32	7.94	87.92	4.72	80.37	5.07	85.87	5.91
+Dice	89.92	7.82	88.44	4.60	80.66	4.98	86.34	5.80
+IoU	90.58	7.78	88.89	4.54	81.89	4.81	87.12	5.71
+Focal	90.66	7.73	89.03	4.51	82.47	4.80	87.39	5.68

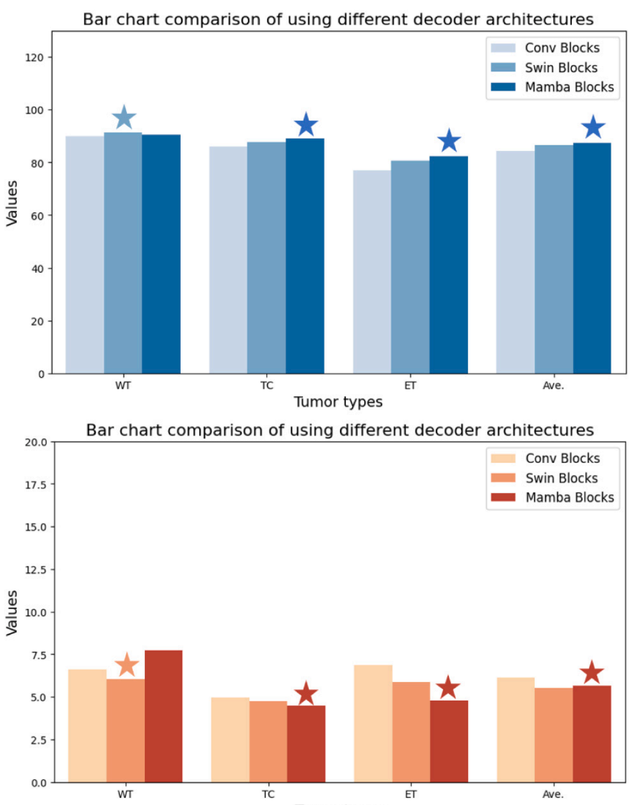

Fig. 7. Dice scores and Haussdorf distance barplots of using different decoder.

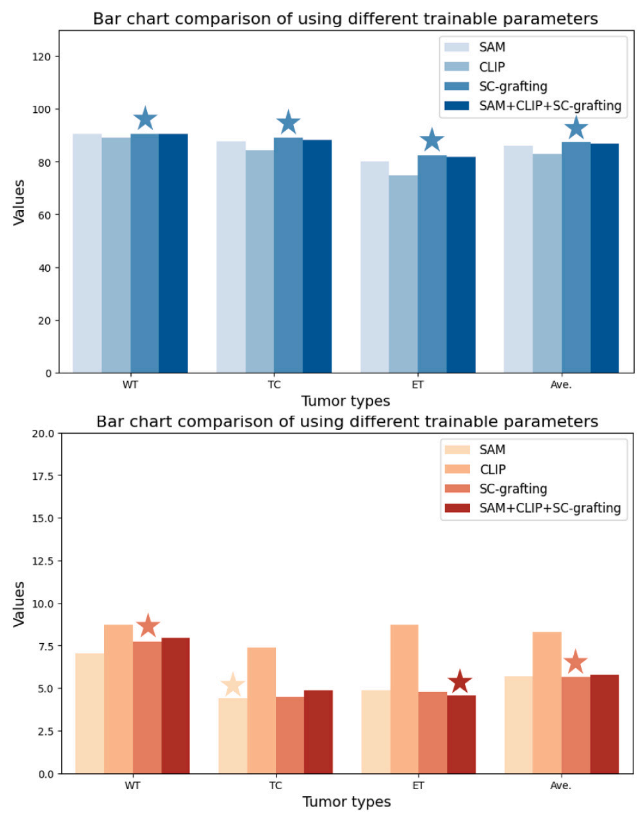

Fig. 8. Dice scores and Haussdorf distance barplots of using different trainable parameters.

other medical CT datasets, we conducted additional ablation experiments to demonstrate the effectiveness of fine-tuning these parameters. As shown in Table 5 and Fig. 8, fine-tuning the parameters of the SAM and CLIP models results in poorer outcomes. Similarly, fine-tuning SAM and CLIP via adapter-based methods also fails to achieve satisfactory results. This is mainly because the capacity of adapters is limited, while SAM is highly sensitive to structural details and CLIP focuses on semantic understanding. Such a lightweight tuning mechanism is insufficient to simultaneously capture both aspects effectively. This may be due to the limited number of training images in the BraTS dataset. Furthermore, based on the results of fine-tuning the SAM, CLIP, and SC-grafting modules, we can infer that the lack of training data may negatively impact the performance of the pretrained SAM and CLIP models. Therefore, we chose to train our SC-grafting module independently for optimal performance.

Effectiveness of dual encoder branch and SC-grafting module. To further validate the effectiveness of our dual encoder design and the SC-grafting module, we visualize the feature spaces of SAM and CLIP using t-SNE before and after fusion. As shown in left of Fig. 9, features extracted from the pretrained SAM after a single STrans layer demonstrate partial overlap with CLIP-derived features, but also exhibit significant redundancy and modality-specific discrepancies. This observation implies limited alignment and inefficient complementarity when directly using the two encoders in parallel.

After applying our proposed SC-grafting mechanism, the resulting features (the right of Fig. 9) demonstrate improved focus on the distinct

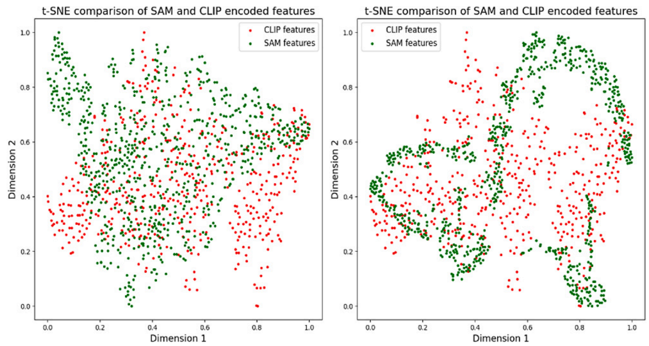

Fig. 9. t-SNE visualization of feature embeddings. (a) Features from SAM encoder show redundancy and partial overlap with CLIP. (b) After SC-grafting, fused features are more compact and complementary, highlighting effective semantic alignment.

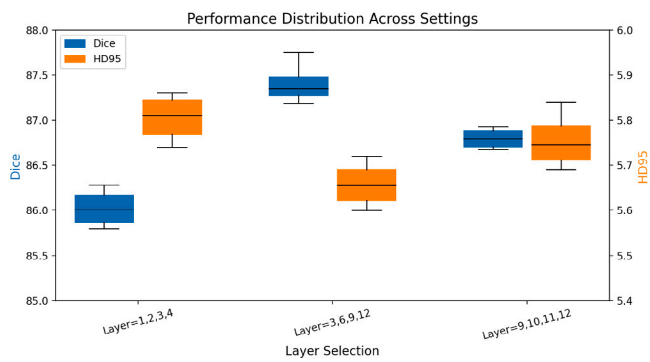

Fig. 10. Performance comparison of Dice coefficient and HD95 distance across different SC-grafting layer configurations. Each box shows the metric distribution over repeated experiments.

and complementary regions of SAM and CLIP. The fused representation emphasizes shared semantics while suppressing redundant components, thus alleviating cross-modal conflict and redundancy. This confirms that SC-grafting effectively enhances feature alignment and contributes to more discriminative joint embeddings.

To investigate the impact of different layer selections for SC-grafting, we conducted an ablation study comparing shallow layers (e.g., Layer 1,2,3,4), deep layers (e.g., Layer 9,10,11,12), and a hybrid multi-level combination (e.g., Layer 3,6,9,12). As shown in Fig. 10, using only shallow features provides fine-grained local details but lacks global context modeling. Conversely, deep-layer features encode semantic abstraction but lose spatial resolution. Both single-sided approaches lead to sub-optimal results. Therefore, we adopt a cross-layer SC-grafting strategy to effectively integrate both detailed and semantic information.

This design aligns with the hierarchical fusion principle widely adopted in segmentation networks, where features from different depths are complementary and collectively enhance model performance.

5. Conclusion

In this paper, we propose a Medical SAM-CLIP Grafting Network for the brain tumor segmentation task. It utilizes a combined SAM-CLIP Dual Encoder with fixed pretrained weights to ensure basic segmentation capability. Additionally, we introduce an SC-grafting module to effectively graft the features of the two models at both resolution scale and semantic dimension. Finally, we employ a Fusion Decoder, which uses Mamba blocks and is assisted by Detail Flow and Semantic Flow, to generate robust brain tumor segmentation results. Experiments have shown competitive results for our proposed method.

CRediT authorship contribution statement

Xinjun Yu: Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Zhoushan Feng:** Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Xiaohong Wu:** Writing – original draft, Investigation, Formal analysis, Data curation. **Jianqiu Chen:** Resources, Project administration. **Weidong Chen:** Visualization, Resources, Project administration. **Baisheng Li:** Writing – review & editing, Validation, Resources, Project administration. **Huan Kuang:** Writing – review & editing, Visualization, Project administration, Funding acquisition, Conceptualization.

Ethics statement

This study was conducted using publicly available datasets, and no new data were collected from human or animal subjects. All datasets used in this research were obtained from publicly accessible sources, and their usage complies with the terms and conditions specified by the original authors or data providers. Additionally, all relevant studies and sources have been appropriately cited in accordance with academic standards. As this study does not involve direct experimentation on human or animal subjects, no ethical approval was required. Furthermore, no personally identifiable information was used, ensuring compliance with privacy and ethical research standards.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yu Xinjun reports financial support was provided by Neurosurgery department, The Sixth Affiliated Hospital of South China, University of Technology, Foshan City, CN. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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